

ME303 Introduction to Mechanical Design

Lecture 02 Materials

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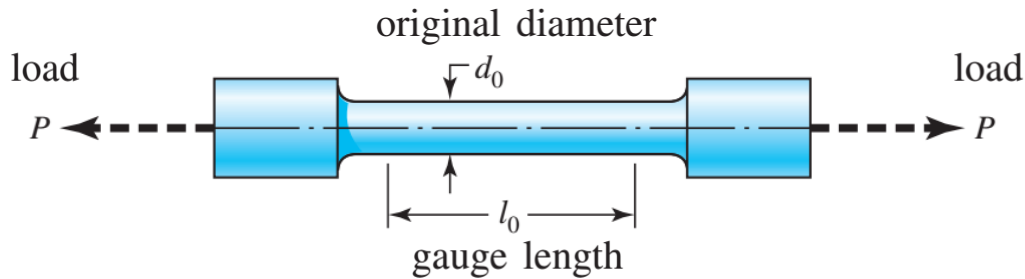
Agenda

Week 01, Friday, Sep 06, 2018

- Material Properties
 - Material Strength and Stiffness
 - Hardness
 - Impact Properties
 - Temperature Effects
- Material Treatments
 - Sand Casting
 - Shell Molding
 - Investment Casting
 - Powder-Metallurgy Process
 - Hot-Working Processes
 - Cold-Working Processes
- Metals
 - The Heat Treatment of Steel
 - Alloy Steels
 - Alloy Steels
 - Corrosion-Resistant Steels
 - Casting Materials
- Nonmetals
 - Polymers
 - Ceramics
 - Composites
- Material Selection

Material Strength & Stiffness

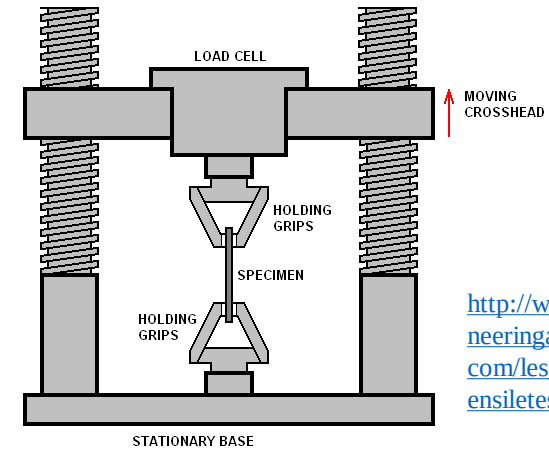
The Tensile Test



stress $\sigma = \frac{P}{A_0}$

original area of the specimen. $A_0 = \frac{1}{4}\pi d_0^2$

normal strain $\epsilon = \frac{l - l_0}{l_0}$



http://www.engineeringarchives.com/les_mom_t_ensiletest.html



http://www.ptli.com/video/t_ensile.jpg

Stress-Strain Diagram

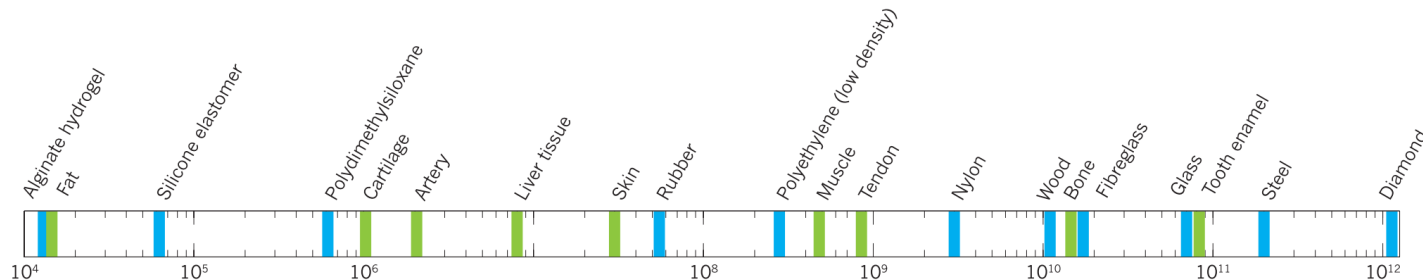
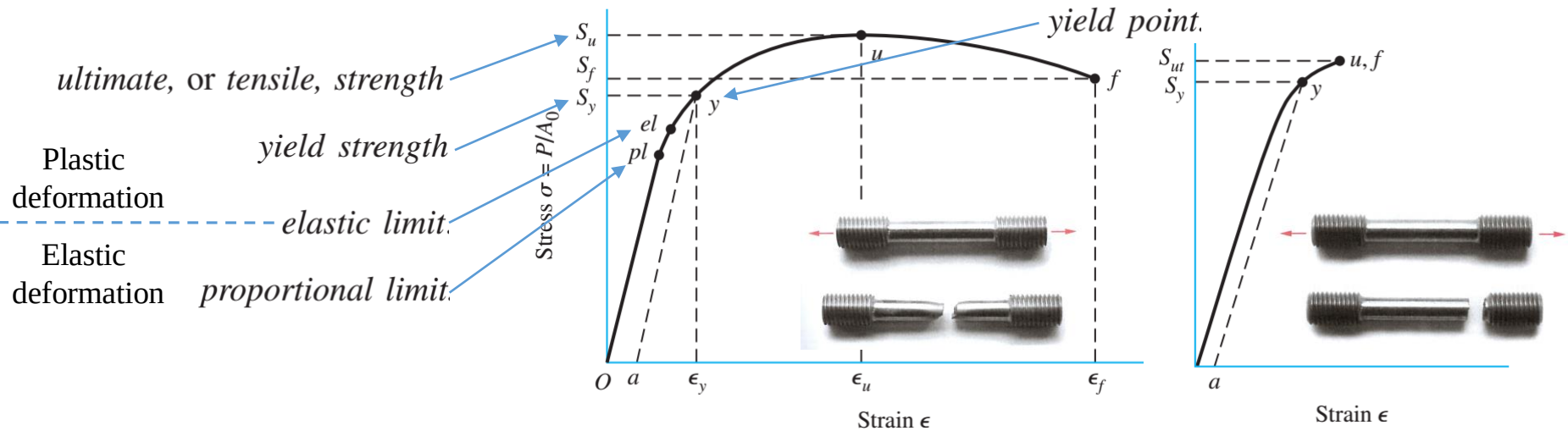
Ductile vs. Brittle Materials

$$\sigma = E\epsilon$$

Young's modulus

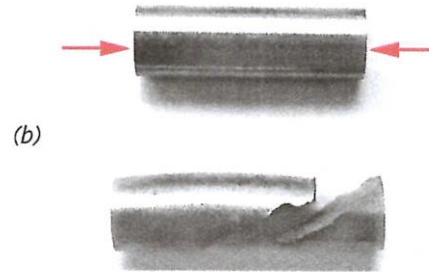
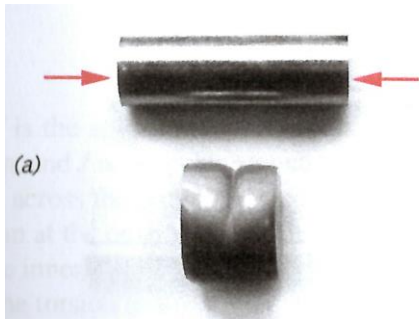
a measure of the stiffness of a material

Engineering Diagram
VS.
True Diagram



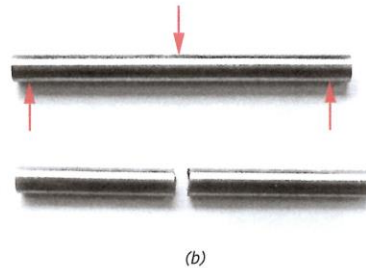
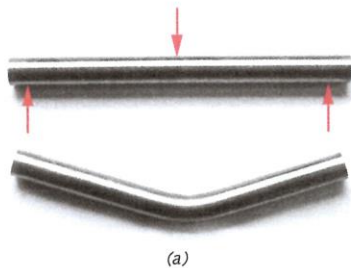
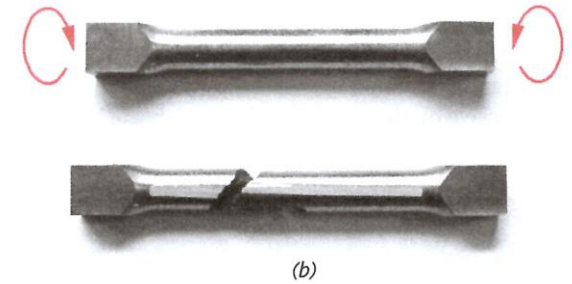
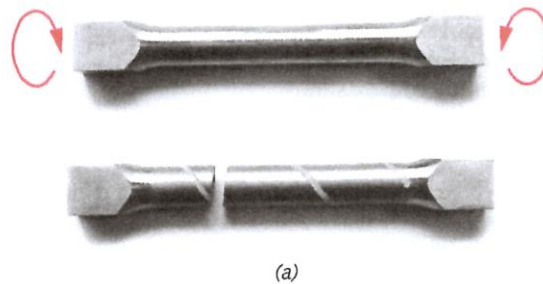
Other Common Material Tests

Ductile vs. Brittle Materials



Compression Test

Torsion Test



Bending Test

Resilience & Toughness

Energy-absorbing Characteristics of a Material

resilience. The capacity of a material to absorb energy within its elastic range
modulus of resilience the energy absorbed per unit volume without permanent deformation.

$$u_R \approx \int_0^{\epsilon_y} \sigma d\epsilon$$

elastic limit is often approximated by the yield point, more readily determined.
 the area under the stress-strain curve up to the elastic limit.

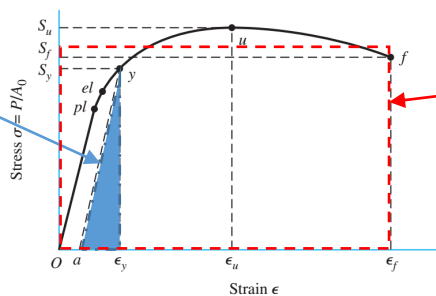
toughness. The capacity of a material to absorb energy without fracture
modulus of toughness the energy absorbed per unit volume without fracture.

$$u_T = \int_0^{\epsilon_f} \sigma d\epsilon$$

the strain at the fracture point.
 the total area under the stress-strain curve up to the fracture point.

$$u_R \approx \frac{1}{2} S_y \epsilon_y = \frac{1}{2} (S_y) (S_y/E) = \frac{S_y^2}{2E}$$

the less stiff material (lower E),
 a greater resilience.



$$u_T \approx \left(\frac{S_y + S_{ut}}{2} \right) \epsilon_f$$

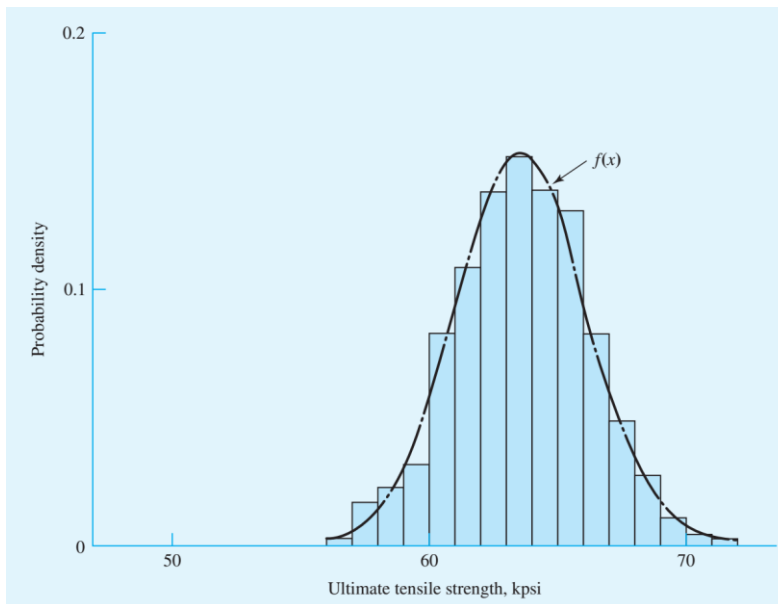
The units of toughness and resilience
 energy per unit volume
 J/m^3

Statistical Significance of Material Properties

Conclusion drawn from one specimen, or from many, many more?

One thousand specimens of 1020 steel were tested to rupture, and the ultimate tensile strengths were reported as follows:

Range midpoint S_{ut} (kpsi)	56.5	57.5	58.5	59.5	60.5	61.5	62.5	63.5	64.5	65.5	66.5	67.5	68.5	69.5	70.5	71.5
Frequency, f_i	2	18	23	31	83	109	138	151	139	130	82	49	28	11	4	2



Testing is an involved and expensive process.

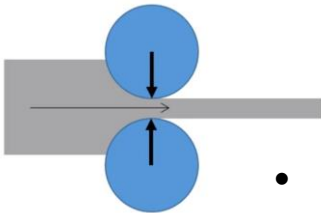
Strength & Cold Work

Strain-hardened

The process of plastic straining below the recrystallization temperature in the plastic region of the stress-strain diagram.

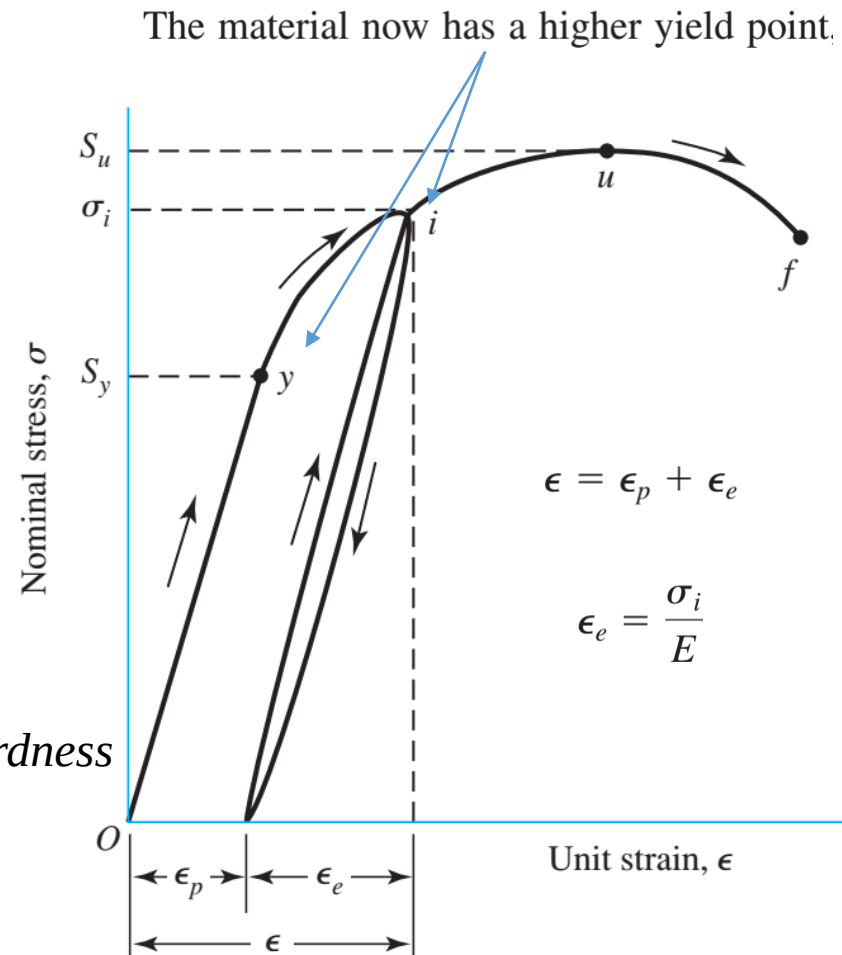
Hot Rolling
Process

- *Wider Tolerance*
- *Lower Strength*
- *Less Expensive*



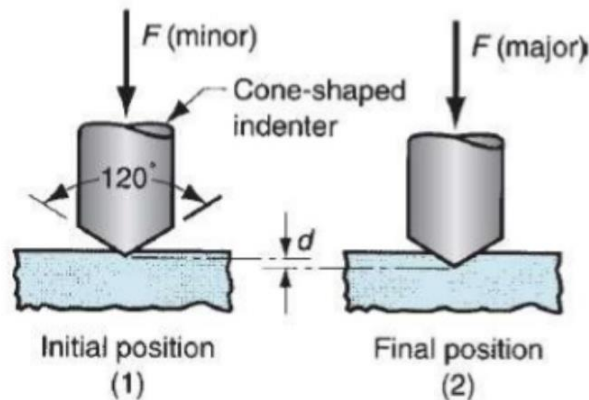
Cold Rolling
Process

- *Tighter Tolerance*
- *Higher Strength & Hardness*
- *More Expensive*
- *More Processing*
- *Better Finish*

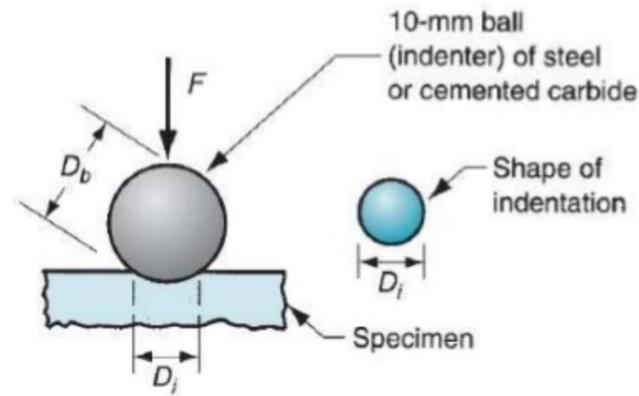


Hardness

The resistance of a material to penetration by a pointed tool.

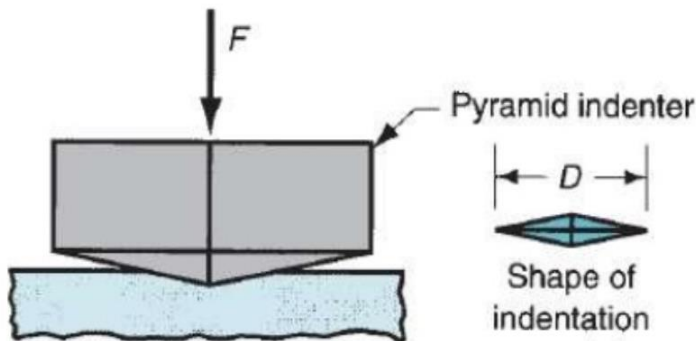


Rockwell

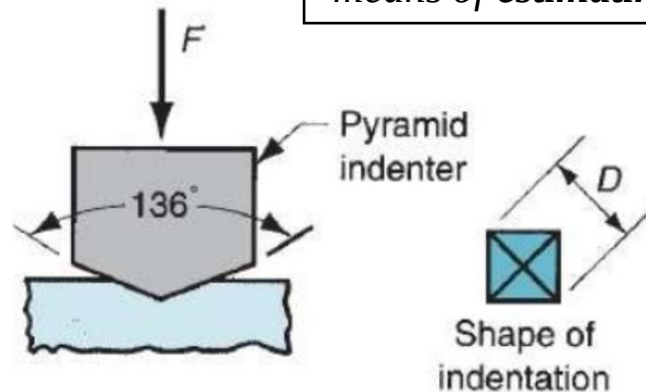


Brinell

*A convenient and nondestructive means of **estimating** material strength.*



Knoop

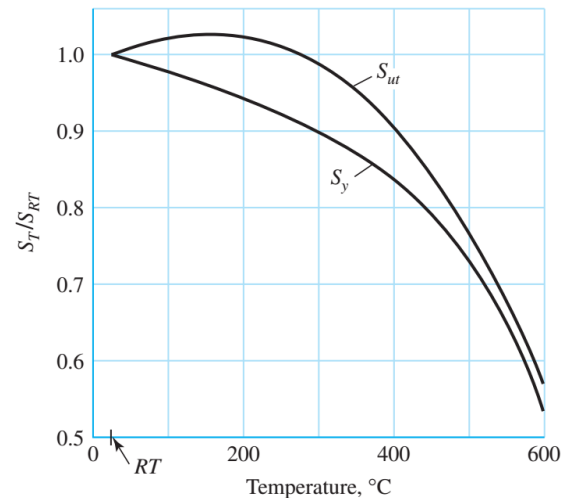
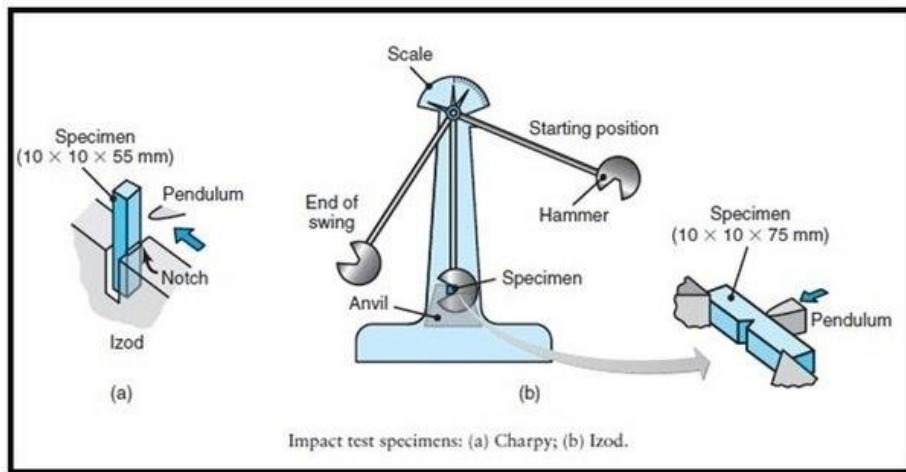
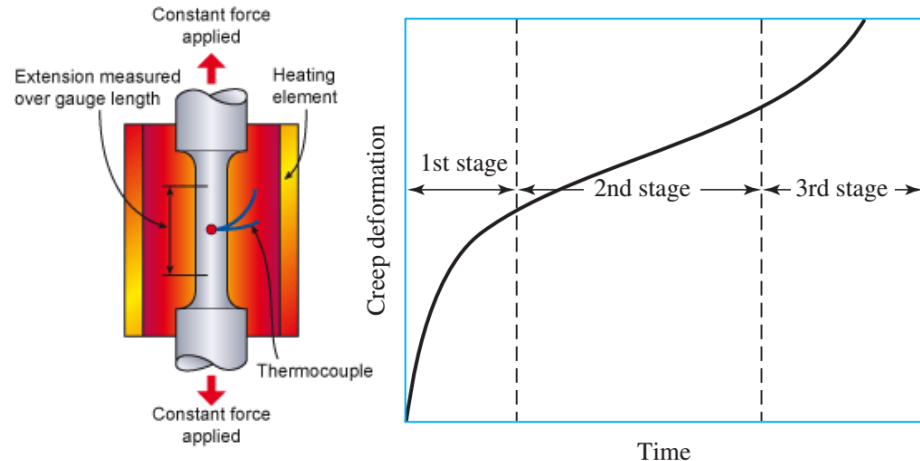


Vickers

Impact Properties & Temperature Effects

Charpy, Izod and Creep Tests

- **[Impact]** if the time of loading is less than one-third the lowest natural period of vibration of the part or structure.
 - Toughness data under dynamic conditions

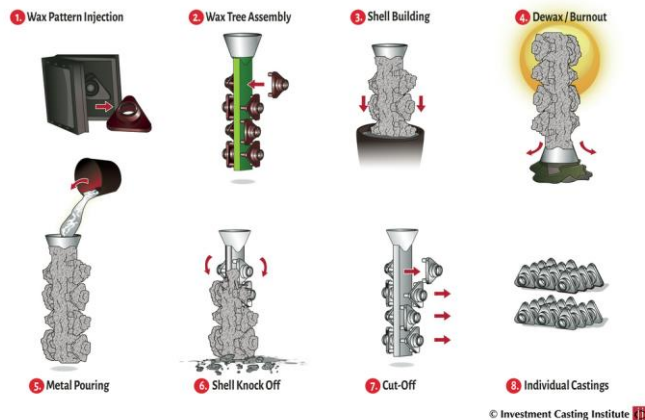
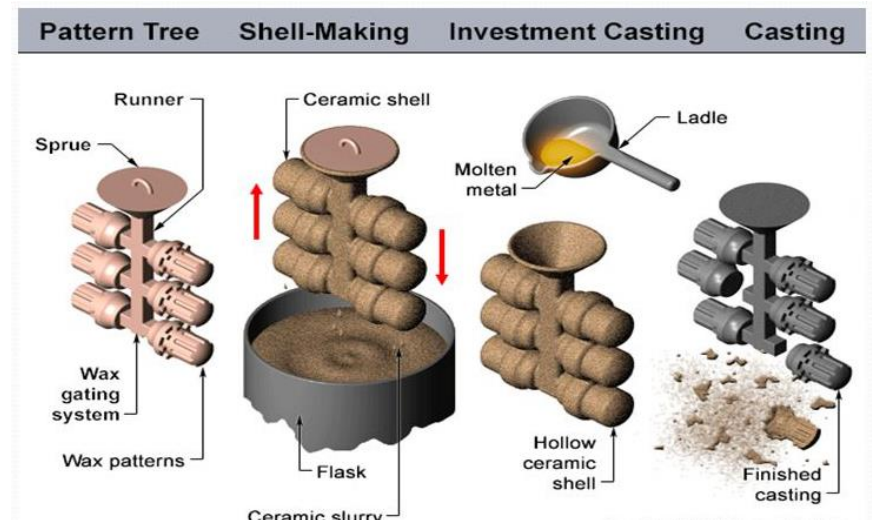


Strength vs.
Temperature

Metal Casting

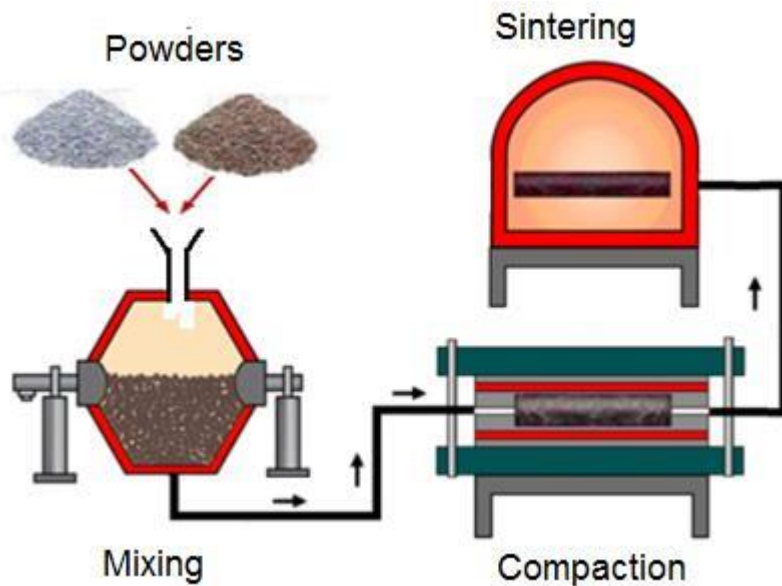
From liquid metal to intended, usually complex, forms using molds.

Sand Casting



Powder-Metallurgy Process

a **quantity-production process** that uses powders from a single metal, several metals, or a mixture of metals and nonmetals.



Advantages:

- (1) elimination of scrap or waste material,
- (2) elimination of machining operations,
- (3) low unit cost when mass-produced,
- (4) exact control of composition.

Disadvantages:

- (1) high cost of dies,
- (2) lower physical properties,
- (3) higher cost of materials,
- (4) limitations on the design,
- (5) limited range of materials that can be used.

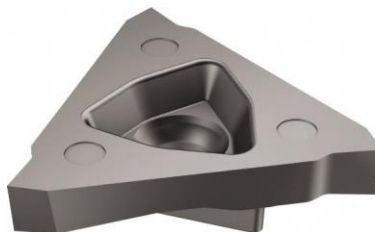
oil-impregnated
bearings



incandescent
lamp filaments



cemented-carbide tips for tools



permanent magnets



surgical implants



Heat Treatment of Steel

time- and temperature-controlled processes that relieve residual stresses and/or modifies material properties such as hardness (strength), ductility, and toughness.

- Annealing:
 - *It involves heating a material above its recrystallization temperature, maintaining a suitable temperature for a suitable amount of time, and then cooling.*
- Quenching:
 - *a process of cooling a metal at a rapid rate.*
- Tempering:
 - *heating steel below the lower critical temperature to impart some toughness*
- Case Hardening
 - *Increase the carbon content at the surface*

Other Metal Materials

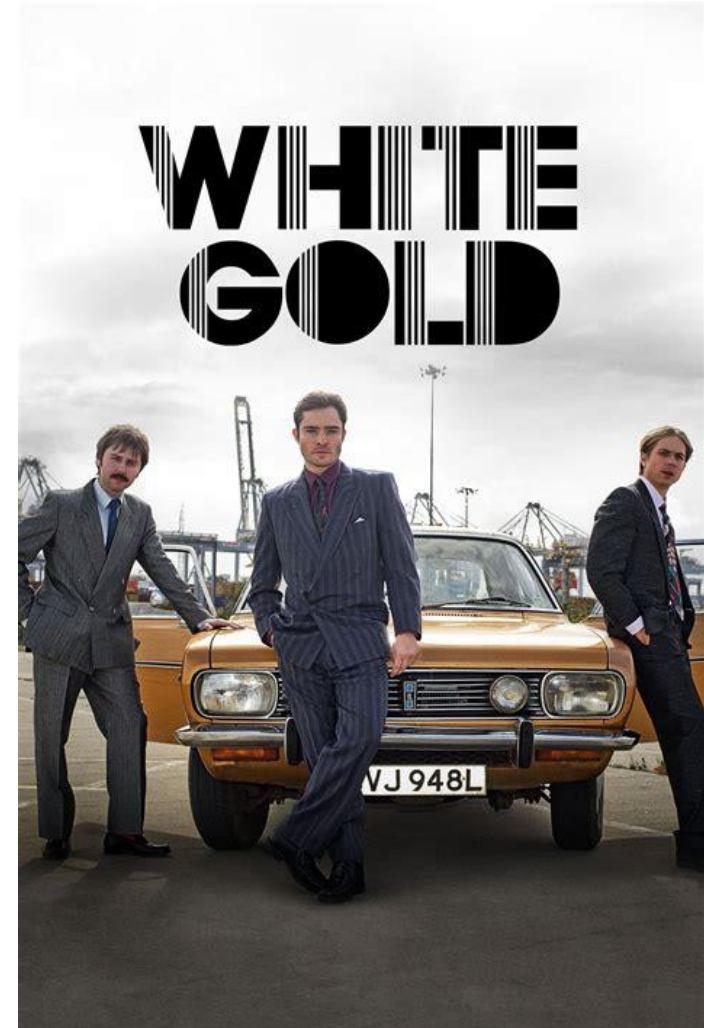
Please refer to the textbook or other Material Courses for more

- Alloy Steels
 - When one or more elements other than carbon are introduced
 - *In sufficient quantities to modify its properties substantially.*
 - Chromium, Nickel, Manganese, Silicon, Molybdenum, Vanadium, Tungsten ...
- Corrosion-Resistant Steels
 - Iron-base alloys containing at least 12 percent chromium
 - *Resistance to many, but not all, corrosive conditions.*
 - the ferritic chromium steels, the austenitic chromium-nickel steels, the martensitic and precipitation-hardenable stainless steels.
- Casting Materials
 - Gray Cast Iron, Ductile and Nodular Cast Iron, White Cast Iron, Malleable Cast Iron, Alloy Cast Irons, Cast Steels, ...

Aluminum and its Alloys

Nonferrous Metals

- Advantages:
 - Strength-weight ratio
 - Resistance to corrosion
 - High thermal and electrical conductivity
 - Cost-versatile for fabrication
- Common Alloying Elements
 - Copper, silicon, manganese, magnesium, and zinc.
- Other nonferrous metals
 - Magnesium, Titanium, Copper-based Alloys



(Thermo)Plastics

Any plastic that flows or is moldable when heat is applied to it.

Name	S_u kpsi	E Mpsi	Hardness Rockwell	Elongation %	Dimensional Stability	Heat Resistance	Chemical Resistance	Processing
ABS group	2–8	0.10–0.37	60–110R	3–50	Good	*	Fair	EMST
Acetal group	8–10	0.41–0.52	80–94M	40–60	Excellent	Good	High	M
Acrylic	5–10	0.20–0.47	92–110M	3–75	High	*	Fair	EMS
Fluoroplastic group	0.50–7	. . .	50–80D	100–300	High	Excellent	Excellent	MPR [†]
Nylon	8–14	0.18–0.45	112–120R	10–200	Poor	Poor	Good	CEM
Phenylene oxide	7–18	0.35–0.92	115R, 106L	5–60	Excellent	Good	Fair	EFM
Polycarbonate	8–16	0.34–0.86	62–91M	10–125	Excellent	Excellent	Fair	EMS
Polyester	8–18	0.28–1.6	65–90M	1–300	Excellent	Poor	Excellent	CLMR
Polyimide	6–50	. . .	88–120M	Very low	Excellent	Excellent	Excellent [†]	CLMP
Polyphenylene sulfide	14–19	0.11	122R	1.0	Good	Excellent	Excellent	M
Polystyrene group	1.5–12	0.14–0.60	10–90M	0.5–60	. . .	Poor	Poor	EM
Polysulfone	10	0.36	120R	50–100	Excellent	Excellent	Excellent [†]	EFM
Polyvinyl chloride	1.5–7.5	0.35–0.60	65–85D	40–450	. . .	Poor	Poor	EFM

*Heat-resistant grades available.

[†]With exceptions.

AncoraS C Coatings L Laminates R Resins E Extrusions M Moldings S Sheet F Foams P Press and sinter methods T Tubing



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Thermosets

A thermoset is a plastic for which the polymerization process is finished in a hot molding press where the plastic is liquefied under pressure.

Cannot be remolded

Name	S_u kpsi	E Mpsi	Hardness Rockwell	Elongation %	Dimensional Stability	Heat Resistance	Chemical Resistance	Processing
Alkyd	3–9	0.05–0.30	99M*	...	Excellent	Good	Fair	M
Allylic	4–10	...	105–120M	...	Excellent	Excellent	Excellent	CM
Amino group	5–8	0.13–0.24	110–120M	0.30–0.90	Good	Excellent*	Excellent*	LR
Epoxy	5–20	0.03–0.30*	80–120M	1–10	Excellent	Excellent	Excellent	CMR
Phenolics	5–9	0.10–0.25	70–95E	...	Excellent	Excellent	Good	EMR
Silicones	5–6	...	80–90M	...	...	Excellent	Excellent	CLMR

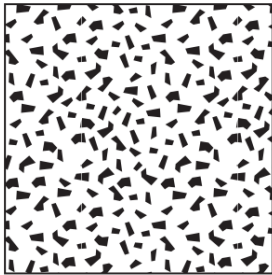
*With exceptions.

C Coatings L Laminates R Resins E Extrusions M Moldings S Sheet F Foams P Press and sinter methods T Tubing

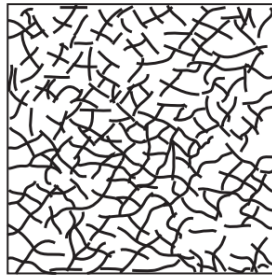
Composite Materials

Please refer to more advanced course for more.

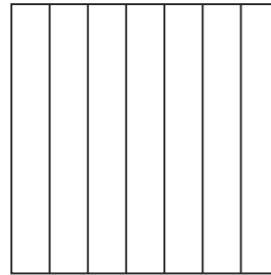
- Formed from two or more dissimilar materials, each of which contributes to the final properties.
 - A Filler (reinforcement) and a Matrix
 - The materials in a composite remain distinct from each other at the macroscopic level.



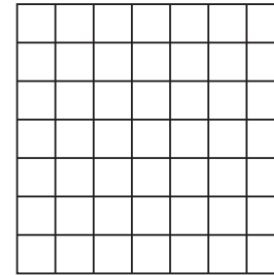
Particulate composite



Randomly oriented short fiber composite



Unidirectional continuous fiber composite



Woven fabric composite

Materials Selection

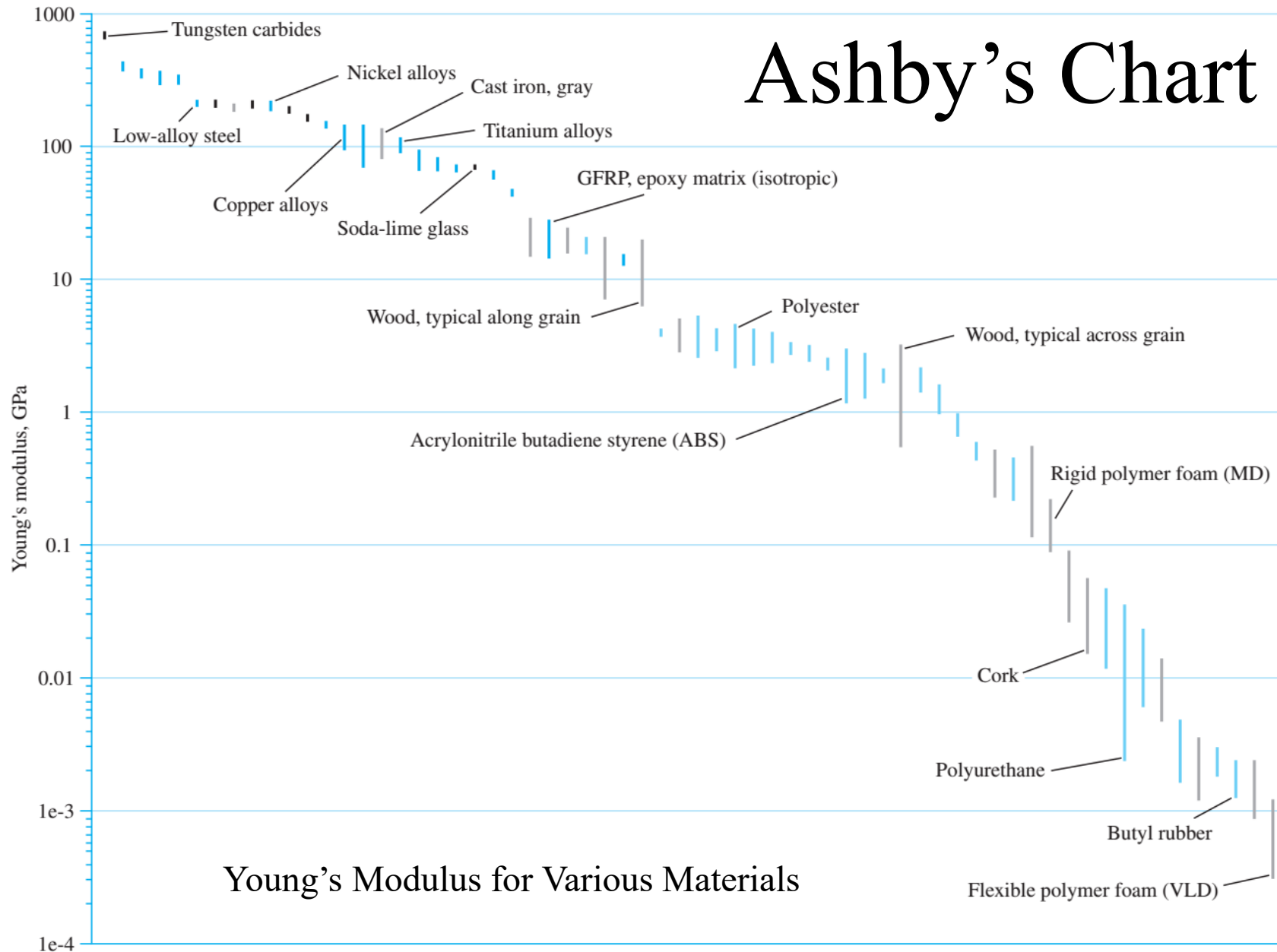
Material Families and Classes

Family	Classes	Short Name
Metals (the metals and alloys of engineering)	Aluminum alloys Copper alloys Lead alloys Magnesium alloys Nickel alloys Carbon steels Stainless steels Tin alloys Titanium alloys Tungsten alloys Lead alloys Zinc alloys	Al alloys Cu alloys Lead alloys Mg alloys Ni alloys Steels Stainless steels Tin alloys Ti alloys W alloys Pb alloys Zn alloys
Ceramics Technical ceramics (fine ceramics capable of load-bearing application) Nontechnical ceramics (porous ceramics of construction)	Alumina Aluminum nitride Boron carbide Silicon carbide Silicon nitride Tungsten carbide Brick Concrete Stone	Al ₂ O ₃ AlN B ₄ C SiC Si ₃ N ₄ WC Brick Concrete Stone
Glasses	Soda-lime glass Borosilicate glass Silica glass Glass ceramic	Soda-lime glass Borosilicate glass Silica glass Glass ceramic
Polymers (the thermoplastics and thermosets of engineering)	Acrylonitrile butadiene styrene Cellulose polymers Ionomers Epoxies Phenolics Polyamides (nylons) Polycarbonate	ABS CA Ionomers Epoxy Phenolics PA PC

Family	Classes	Short Name
Polymers (<i>continued</i>)	Polyesters Polyetheretherketone Polyethylene Polyethylene terephthalate Polymethylmethacrylate Polyoxymethylene(Acetal) Polypropylene Polystyrene Polytetrafluorethylene Polyvinylchloride	Polyester PEEK PE PET or PETE PMMA POM PP PS PTFE PVC
Elastomers (engineering rubbers, natural and synthetic)	Butyl rubber EVA Isoprene Natural rubber Polychloroprene (Neoprene) Polyurethane Silicon elastomers	Butyl rubber EVA Isoprene Natural rubber Neoprene PU Silicones
Hybrids Composites Foams Natural materials	Carbon-fiber reinforced polymers Glass-fiber reinforced polymers SiC reinforced aluminum Flexible polymer foams Rigid polymer foams Cork Bamboo Wood	CFRP GFRP Al-SiC Flexible foams Rigid foams Cork Bamboo Wood

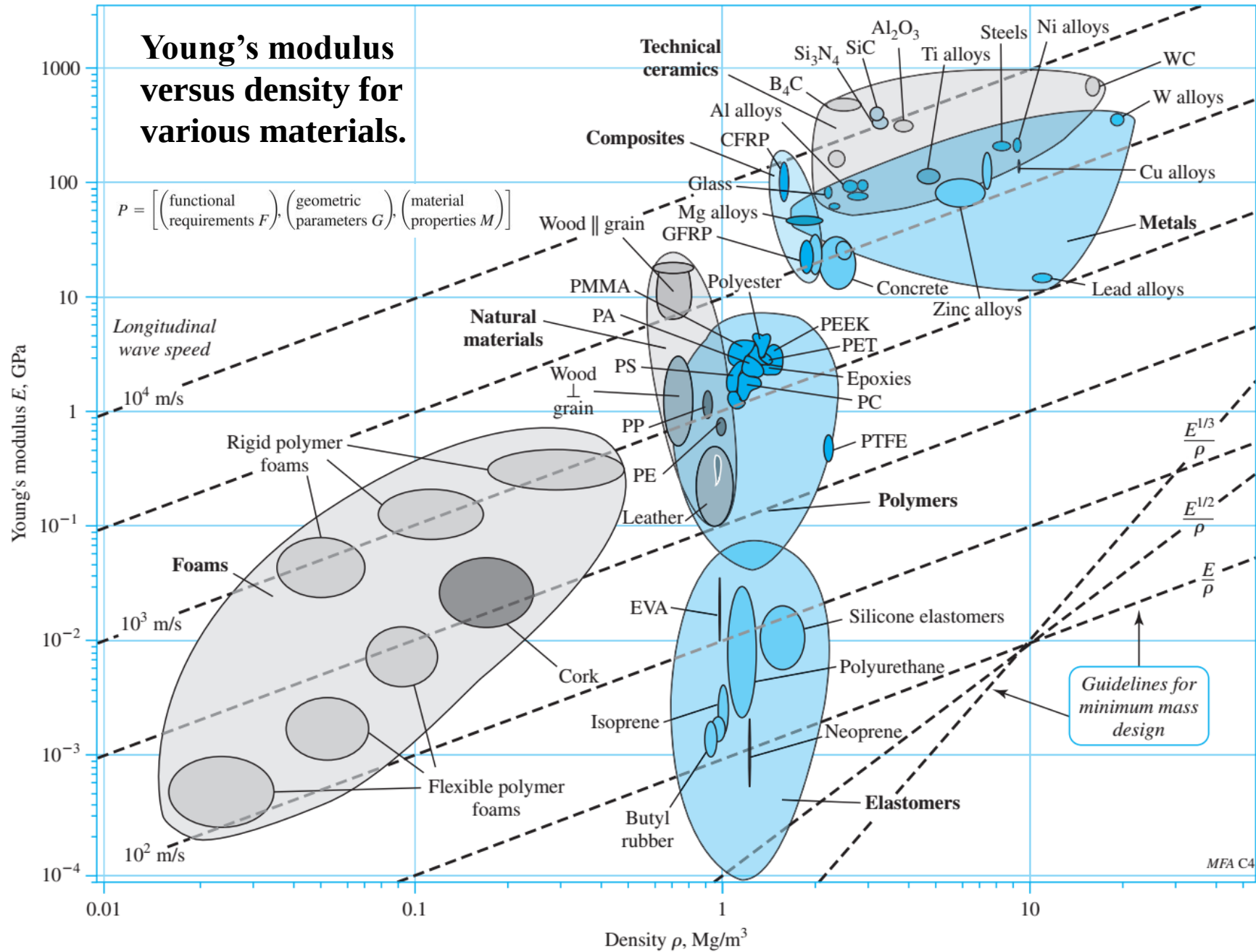
From M. F. Ashby, *Materials Selection in Mechanical Design*, 3rd ed., Elsevier Butterworth-Heinemann, Oxford, 2005. Table 4–1, pp. 49–50.

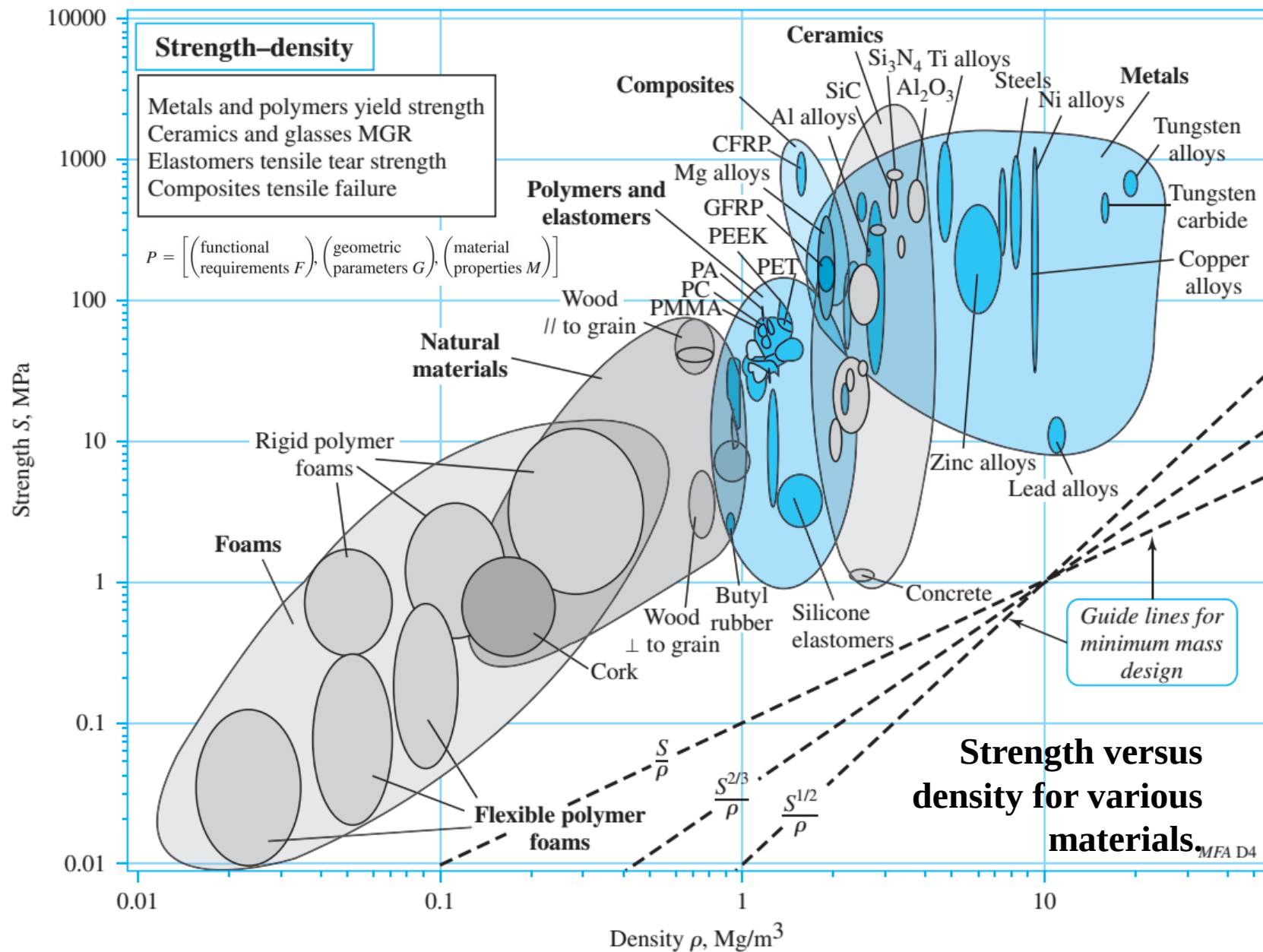
Ashby's Chart



Young's modulus versus density for various materials.

$$P = \left[\begin{pmatrix} \text{functional} \\ \text{requirements } F \end{pmatrix}, \begin{pmatrix} \text{geometric} \\ \text{parameters } G \end{pmatrix}, \begin{pmatrix} \text{material} \\ \text{properties } M \end{pmatrix} \right]$$





Next class

- **Lecture Topic:** Kinematics & Load Determination
- Wednesday 1400-1600
- Room 206, 2 Lychee Park

Thank you!

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